

CSC 345 - Project #4: AI Use Essay

Throughout the completion of Project 4, we used artificial intelligence to gain a better understanding of the material and the expectations of the assignment. Socket programming is a challenging concept which artificial intelligence was able to simplify to be more understandable. The type of artificial intelligence we specifically used was Claude.

When beginning the project, we found as many resources as possible, whether it was on GitHub, GeeksForGeeks, or various Linux forums. This helped quite a bit with the project and helped me get the first four requirements completed and working. When we ran into difficulties, we used Claude to further explain concepts to implement them into the project.

Another use we had for artificial intelligence was for debugging. This was incredibly helpful, as we had several small errors that would've been difficult to catch checking line-by-line. When implementing the various checkpoints required by `main_server.c` and `main_client.c`, there were small errors Claude helped me catch. One example was several times I missed semicolons at the end of a call. Or there were several times I made a typo such as when I called `colors_assigned` as `colors_assogned`, but everywhere else it was spelled correctly. Claude helped me find these errors that caused issues in the output that would've taken me much longer to catch on my own.

Furthermore, we would ask specific questions about why things weren't working. We'd only take this path after looking at forums and websites online. Using artificial intelligence for this reason was also very helpful and explained what was wrong in a way that still allowed me to solve the problem myself, and learn for the future.

Overall, I found using artificial intelligence helpful in completing this project, but not to the point where it was necessary. I especially liked using Claude, as I feel it allows the user to not rely on artificial intelligence but truly use it as a tool. It helped me organize my workflow and give me useful information to solve problems, but not to solve the problems for me.